

Name (Print): _____

Score: / 345

1. 15 points Let $f(x) = 8x^3 - 12x^2 + 6x - 1$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[0, 1]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.
2. 15 points Let $f(x) = x^2 - 5$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[-2, 2]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

3. 15 points Let $f(x) = x^2 - 5$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[1, 5]$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.
4. 15 points Let $f(x) = x^2 - 4x + 8$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $[0, \infty)$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

5. 15 points Let $f(x) = x - \frac{2}{x} + \frac{5}{x^3}$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $(0, \infty)$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.
6. 15 points (bonus) Let $f(x) = \frac{x^6}{e^x}$. Find the absolute minimum and the absolute maximum of $f(x)$ over the interval $(0, \infty)$, if such an absolute maximum or minimum exists. For every absolute extremum, give the point $(x, f(x))$ and indicate whether it is a minimum or maximum.

Section 12.6

1. 20 points A company manufactures and sells x cellphones per week. The weekly price-demand and cost equations are given below.

$$p = 600 - 0.5x \text{ and } C(x) = 15,000 + 140x.$$

What price should the company charge for the phones, and how many phones should be produced, to maximize the weekly revenue? What is the maximum weekly revenue?

2. 20 points A box is to be made at a manufacturing company out of a rectangular piece of cardboard 15 inches wide and 20 inches long. Squares of equal size will be cut out of each corner, and then the ends and sides will be folded up to form a rectangular box with an open top. What size square should be cut from each corner to maximize volume?

3. 20 points A deli sells 720 sandwiches per day at a price of \$6 each. A market survey shows that for every \$0.10 reduction in the price, 40 more sandwiches will be sold. How much should the deli charge in order to maximize revenue?

Section 13.1

1. 10 points Find $\int \frac{6}{x} dx$

2. 10 points Find f such that $f'(x) = \frac{5}{\sqrt{x}}$ and $f(16) = 55$.

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3. 10 points Find y such that $\frac{dy}{dt} = 5e^t - 4$ and $y(0) = 4$.
4. 10 points If the marginal profit of producing x surfboards is given by $P'(x) = 40x - x^2$, and $P(10) = 20$ where $P(x)$ is the profit in dollars, find $P(x)$.

5. 10 points Find $\int \frac{x^3 e^x - 3x^2}{x^3} dx$.

6. 15 points (bonus) Find $\int \frac{d}{dx}(x^4 + 5x^2 + 8) dx$.

Section 13.2

1. In each problem, find $\int f(x) dx$.

(a) 10 points $f(x) = 2x(x^2 + 2)^{11}$

(b) 10 points $f(x) = 10e^{10x}$

(c) 10 points $f(x) = x^{12}e^{x^{13}}$

(d) 10 points $f(x) = \frac{x^4}{(6 - x^5)^2}$

2. 10 points Find $\int x\sqrt{12 - x^2}dx$.

3. 10 points If the marginal cost for producing x of a certain item is given by $C'(x) = 8(x+10)^2$, and the fixed cost is $C(0) = 2,000$ (in dollars), find the cost function $C(x)$.

Section 13.4

1. 10 points Calculate $\int_4^9 9x + x^2 dx$, given that
- $$\int_4^9 x^2 dx = \frac{665}{3}, \text{ and } \int_4^9 x dx = \frac{65}{2}.$$

2. 10 points Calculate $\int_1^{10} 5x^2 + x dx$ given that
- $$\int_1^9 x^2 dx = \frac{728}{3}, \int_9^{10} x^2 dx = \frac{271}{3}, \text{ and } \int_1^{10} x dx = \frac{99}{2}.$$

3. 10 points Let $f(x) = x^2 + 1$. Over the interval $[0, 4]$, calculate R_4 , the right-hand Riemann sum.

4. 10 points Let $f(x) = x^2 + 1$. Over the interval $[0, 4]$, calculate L_4 , the left-hand Riemann sum.

Section 13.5

1. 10 points Calculate $\int_2^3 6x^2(x^3 + 2)^{-2} \, dx$.

2. 10 points Calculate $\int_1^{10} 3x^2 + 2x + 1 \, dx$.

3. 10 points Calculate $\int_0^3 2e^{2x} \, dx$.

4. 10 points Calculate $\int_{-2}^2 x^3 \, dx$.

5. 10 points Calculate the average value of $f(x) = 5 - x^2$ over the interval $[-3, 2]$.

6. 10 points Calculate the average value of $f(x) = 32e^{0.04x}$ over the interval $[10, 30]$.